



US 20250275909A1

(19) **United States**

(12) **Patent Application Publication**
Tanaka

(10) **Pub. No.: US 2025/0275909 A1**

(43) **Pub. Date: Sep. 4, 2025**

(54) **COSMETIC COMPOSITION COMPRISING
MICA AND LAUROYL LYSINE**

A61K 8/44 (2006.01)

A61Q 19/00 (2006.01)

(71) Applicant: **The Procter & Gamble Company,**
Cincinnati, OH (US)

(52) **U.S. Cl.**

CPC *A61K 8/965* (2013.01); *A61K 8/0245*
(2013.01); *A61K 8/0275* (2013.01); *A61K*
8/442 (2013.01); *A61Q 19/00* (2013.01); *A61K*
2800/10 (2013.01); *A61K 2800/28* (2013.01);
A61K 2800/412 (2013.01); *A61K 2800/621*
(2013.01)

(72) Inventor: **Shuhei Tanaka**, Singapore (SG)

(21) Appl. No.: **19/211,462**

(22) Filed: **May 19, 2025**

(57)

ABSTRACT

Related U.S. Application Data

(63) Continuation of application No. 18/081,743, filed on
Dec. 15, 2022.

(60) Provisional application No. 63/352,642, filed on Jun.
16, 2022, provisional application No. 63/291,948,
filed on Dec. 21, 2021.

A cosmetic composition containing mica, lauroyl lysine, and
water. The cosmetic composition meets at least one of the
following conditions (i)-(iii): (i) wherein the weight ratio
between the mica and lauroyl lysine is from about 2:1 to
about 1:10; (ii) wherein the mica has a mean particle size of
from about 5 microns to about 20 microns, and the lauroyl
lysine has a mean particle size of from about 10 microns to
about 40 microns, and the mean particle size ratio between
the mica and lauroyl lysine is from about 1:1 to about 1:5;
and (iii) wherein the lauroyl lysine has a flat polygonal
shape. The cosmetic composition of the present invention
can provide benefits from mica with reduced friction/re-
duced draggy feeling.

Publication Classification

(51) **Int. Cl.**

A61K 8/96 (2006.01)

A61K 8/02 (2006.01)

COSMETIC COMPOSITION COMPRISING MICA AND LAUROYL LYSINE

FIELD OF THE INVENTION

[0001] The present invention is directed to a cosmetic composition comprising by weight: from about 0.01% to about 5% of mica; from about 0.1% to about 10% of lauroyl lysine; and from about 55% to about 99% of water, wherein the composition meets at least one of the following conditions (i)-(iii): (i) wherein the weight ratio between the mica and lauroyl lysine is from about 2:1 to about 1:10; (ii) wherein the mica has a mean particle size of from about 5 microns to about 20 microns, and the lauroyl lysine has a mean particle size of from about 10 microns to about 40 microns, and the mean particle size ratio between the mica and lauroyl lysine is from about 1:1 to about 1:5; and (iii) wherein the lauroyl lysine has a flat polygonal shape. The cosmetic composition of the present invention provides benefits from mica with reduced friction/reduced draggy feeling.

BACKGROUND OF THE INVENTION

[0002] Mammalian keratinous tissue, particularly human skin, is subjected to a variety of insults by both extrinsic and intrinsic factors. Such extrinsic factors include ultraviolet radiation, environmental pollution, wind, heat, infrared radiation, low humidity, harsh surfactants, abrasives, etc. Intrinsic factors, on the other hand, include chronological aging and other biochemical changes from within the skin. Whether extrinsic or intrinsic, these factors result in visible signs of skin damage. Typical skin damages in aging or damaged skin include fine lines, wrinkling, hyperpigmentation, sallowness, sagging, dark under-eye circles, puffy eyes, enlarged pores, diminished rate of turnover, and abnormal desquamation or exfoliation. Additional damage incurred as a result of both external and internal factors includes visible dead skin i.e., flaking, scaling, dryness, and roughness.

[0003] Currently, there are a number of personal care products that are available to consumers, which are directed toward improving the health and physical appearance of keratinous tissues such as the skin, hair, and nails. The majority of these products are directed to delaying, minimizing or even eliminating skin wrinkling, spots, and other histological changes typically associated with the aging of skin or environmental damage to human skin. Consumers may prefer topically applied products since they are not only effective, but also safe and pleasant to use.

[0004] Some of cosmetic products contain interference pigments/pearlescent pigments such as mica for providing, for example, an alternative to colored pigments for diminishing the appearance of imperfections such as spots, pores and fine lines. Such interference pigments/pearlescent pigments typically are thin and plate-like particles.

[0005] However, it has been found by the present inventors that, among such thin and plate-like particles, Mica especially TiO₂ coated mica may provide friction/draggy feeling during the application to skin, especially: when mica is contained in cosmetic compositions containing a higher level of water; when mica is contained in the cosmetic compositions containing lower level of oils; and/or when mica is contained at a higher level in such cosmetic compositions.

[0006] Based on the foregoing, there is a need for a cosmetic composition, which provides benefits from mica with reduced friction/reduced draggy feeling, especially: when mica is contained in cosmetic compositions containing higher level of water; when mica is contained in the cosmetic compositions containing lower level of oils; and/or when mica is contained at a higher level in such cosmetic compositions.

[0007] Additionally, there may exist a need for a cosmetic composition, which provides at least one of the followings:

[0008] benefits from mica with reduced heavy coated feel and/or reduced friction/reduced draggy feeling, while providing fresh/light feel from the composition containing a higher level of water;

[0009] benefits from mica especially radiance and/or brightness from mica with reduced heavy coated feel and/or reduced friction/reduced draggy feeling, while providing reduced greasy oily shine from oil from the composition; and

[0010] improved benefits from mica, especially radiance and/or brightness, by higher level of mica, with reduced friction/reduced draggy feeling.

SUMMARY OF THE INVENTION

[0011] The present invention is directed to a cosmetic composition comprising by weight:

[0012] from about 0.01% to about 5% of mica;

[0013] from about 0.1% to about 10% of lauroyl lysine; and

[0014] from about 55% to about 99% of water;

[0015] wherein the composition meets at least one of the following conditions (i)-(iii):

[0016] (i) wherein the weight ratio between the mica and lauroyl lysine is from about 2:1 to about 1:10;

[0017] (ii) wherein the mica has a mean particle size of from about 5 microns to about 20 microns, and the lauroyl lysine has a mean particle size of from about 10 microns to about 40 microns, and the mean particle size ratio between the mica and lauroyl lysine is from about 1:1 to about 1:5; and

[0018] (iii) wherein the lauroyl lysine has a flat polygonal shape.

[0019] It has now been surprisingly found that, by the use of lauroyl lysine in specific conditions together with mica, the composition of the present invention provides benefits from mica with reduced friction/reduced draggy feeling, even: when mica is contained in cosmetic compositions containing higher level of water; when mica is contained in the cosmetic compositions containing lower level of oils; and/or when mica is contained at a higher level in such cosmetic compositions.

[0020] Thus, the composition of the present invention provides at least one of the followings: benefits from mica with reduced heavy coated feel and/or reduced friction/reduced draggy feeling, while providing fresh/light feel from the composition containing a higher level of water;

[0021] benefits from mica especially radiance and/or brightness from mica with reduced heavy coated feel and/or reduced friction/reduced draggy feeling, while providing reduced greasy oily shine from oil from the composition; and

[0022] improved benefits from mica, especially radiance and/or brightness, by higher level of mica, with reduced friction/reduced draggy feeling.

DETAILED DESCRIPTION OF THE INVENTION

[0023] Reference within the specification to “embodiment (s)” or the like means that a particular material, feature, structure and/or characteristic described in connection with the embodiment is included in at least one embodiment, optionally a number of embodiments, but it does not mean that all embodiments incorporate the material, feature, structure, and/or characteristic described. Furthermore, materials, features, structures and/or characteristics may be combined in any suitable manner across different embodiments, and materials, features, structures and/or characteristics may be omitted or substituted from what is described. Thus, embodiments and aspects described herein may comprise or be combinable with elements or components of other embodiments and/or aspects despite not being expressly exemplified in combination, unless otherwise stated or an incompatibility is stated.

[0024] In all embodiments, all ingredient percentages are based on the weight of the cosmetic composition, unless specifically stated otherwise. All ratios are weight ratios, unless specifically stated otherwise. The number of significant digits conveys neither a limitation on the indicated amounts nor on the accuracy of the measurements. All numerical amounts are understood to be modified by the word “about” unless otherwise specifically indicated. Unless otherwise indicated, all measurements are understood to be made at approximately 25° C. and at ambient conditions, where “ambient conditions” means conditions under about 1 atmosphere of pressure and at about 50% relative humidity. All numeric ranges are inclusive and combinable to form narrower ranges not explicitly disclosed. For example, delineated upper and lower range limits are interchangeable to create further ranges.

[0025] The compositions of the present invention can comprise, consist essentially of, or consist of, the essential components as well as optional ingredients described herein. As used herein, “consisting essentially of” means that the composition or component may only include additional ingredients that do not materially alter the basic and novel characteristics of the claimed composition or method. As used in the description and the appended claims, the singular forms “a”, “an”, and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise.

Definitions

[0026] “About” modifies a particular value by referring to a range equal to plus or minus twenty percent (+/-20%) or less (e.g., less than 15%, 10%, or even less than 5%) of the stated value.

[0027] “Apply” or “application”, as used in reference to a composition, means to apply or spread the compositions of the present invention onto a human skin surface such as the epidermis.

[0028] “Derivative,” herein, means amide, ether, ester, amino, carboxyl, acetyl, and/or alcohol derivatives of a given compound.

[0029] “Effective amount” means an amount of a compound or composition sufficient to significantly induce a positive benefit to keratinous tissue over the course of a treatment period. The positive benefit may be a health, appearance, and/or feel benefit, including, independently or in combination, the benefits disclosed herein.

[0030] “Cosmetic agent” means any substance, as well as any component thereof, intended to be rubbed, poured, sprinkled, sprayed, introduced into, or otherwise applied to a mammalian body or any part thereof to provide a cosmetic effect. Cosmetic agents may include substances that are Generally Recognized as Safe (GRAS) by the US Food and Drug Administration, food additives, and materials used in non-cosmetic consumer products including over-the-counter medications.

[0031] “Cosmetic composition” means a composition comprising a cosmetic agent. Examples of cosmetic compositions include color cosmetics (e.g., foundations, lipsticks, concealers, and mascaras), skin care compositions (e.g., moisturizers and sunscreens), personal care compositions (e.g., rinse-off and leave on body washes and soaps), hair care compositions (e.g., shampoos and conditioners).

[0032] “Skin care” means regulating and/or improving a skin condition (e.g., skin health, appearance, or texture/feel). Some nonlimiting examples of improving a skin condition include improving skin appearance and/or feel by providing a smoother, more even appearance and/or feel; increasing the thickness of one or more layers of the skin; improving the elasticity or resiliency of the skin; improving the firmness of the skin; and reducing the oily, shiny, and/or dull appearance of skin, improving the hydration status or moisturization of the skin, improving the appearance of fine lines and/or wrinkles, improving skin exfoliation or desquamation, plumping the skin, improving skin barrier properties, improve skin tone, reducing the appearance of redness or skin blotches, and/or improving the brightness, radiancy, or translucency of skin.

[0033] “Skin care active” means a compound or combination of compounds that, when applied to skin, provide an acute and/or chronic benefit to skin or a type of cell commonly found therein. Skin care actives may regulate and/or improve skin or its associated cells (e.g., improve skin elasticity, hydration, skin barrier function, and/or cell metabolism).

[0034] “Skin care composition” means a composition that includes a skin care active and regulates and/or improves skin condition.

[0035] “Synergy,” and variations thereof, means that the effect provided by a combination of two or more materials is more than the additive effect expected for these materials.

[0036] “Treatment period,” as used herein, means the length of time and/or frequency that a material or composition is applied to a target skin surface.

Cosmetic Composition

[0037] The cosmetic composition of the present invention comprises: mica; lauroyl lysine; and water, wherein the composition meets at least one of the following conditions (i)-(iii):

[0038] (i) wherein the weight ratio between the mica and lauroyl lysine is from about 2:1 to about 1:10, alternatively from about 1:1 to about 1:7, alternatively from about 1:1.2 to about 1:7, alternatively from about 1:1.5 to about 1:5.

[0039] (ii) wherein the mica has a mean particle size of from about 5 microns to about 20 microns, and the lauroyl lysine has a mean particle size of from about 10 microns to about 40 microns, and the mean particle size ratio between the mica and lauroyl lysine is from about

1:1 to about 1:5, alternatively from about 1:1.2 to about 1:4, alternatively from about 1:1.4 to about 1:4; and

[0040] wherein the lauroyl lysine has a flat polygonal shape.

[0041] The composition has a viscosity of from about 100 cP to about 20,000 cP, alternatively from about 1000 cP to about 15,000 cP, especially when the composition has a higher level of water.

[0042] If the composition of the present invention contains Cetyl alcohol, Stearyl alcohol and/or Behenyl alcohol, it may be preferred that the composition contains a limited amount of such Cetyl alcohol, Stearyl alcohol and Behenyl alcohol, for providing fresh/light feel from the composition containing a higher level of water. Total level of cetyl, stearyl, and behenyl alcohols in the composition can be up to about 0.5%, alternatively up to about 0.3%, alternatively up to about 0.1%, and alternatively 0%.

[0043] If the composition contains liquid oils (at ambient) having a higher refractive index, it may be preferred that the composition contains a limited amount of such liquid oils i.e., those having a refractive index of over 1.455, for providing reduced greasy gloss/oily shine from oils from the composition. Total level of such liquid oils in the composition is up to about 3%, alternatively up to about 1%, alternatively up to about 0.5%.

[0044] If the composition contains liquid oils (at ambient), total level of such liquid oils (at ambient) in the composition, regardless of the refractive index, up to about 10%, alternatively 7% or less.

[0045] The cosmetic compositions herein are intended for topical application to human skin. The compositions herein may optionally include one or more additional skin actives or other ingredients of the type commonly included in topical cosmetic compositions.

[0046] The cosmetic compositions herein may be cosmetic compositions, pharmaceutical compositions, or cosmeceutical compositions, and may be provided in various product forms, including, but not limited to, solutions, suspensions, lotions, gels, toners, cleansing liquid washes, hydrogels, film-forming products, and the like. In some instances, the composition form may follow from the particular dermatologically acceptable carrier chosen. For example, the composition (and carrier) may be provided in the form of an emulsion (e.g., water-in-oil, oil-in-water, or water-in-oil-in water) or an aqueous dispersion. The cosmetic composition can be in the form of an oil-in-water emulsion.

Mica

[0047] The composition can comprise mica at a level of from about 0.01% to about 5%, alternatively from about 0.05% to about 3%, alternatively from about 0.1% to about 2% of mica.

[0048] The mica may have a mean particle size of from about 5 microns to about 20 microns, alternatively from about 7 microns to about 17 microns, alternatively from about 10 microns to about 17 microns.

[0049] The mica useful herein can be natural and/or synthetic, for example, synthetic mica may have an INCI name of Fluorophlogopite. The mica useful herein are thin and plate-like particles, and can be a base substrate particle layered with film of TiO₂, with or without additives of tin oxide, wherein the thickness of the layers is from about 50 nm to about 300 nm. Such mica layered with TiO₂ can be

further coated by iron oxide. Mica with or without layered films of TiO₂ etc. has an aspect ratio of at least 5, alternatively more than 10, alternatively more than 20. The aspect ratio can be determined by the average thickness and the average length (also called as average particle size) according to the formula: aspect ratio=length/thickness.

[0050] Mica useful herein may be synthetic mica in view of reducing friction/draggy feel and in view of controlling heavy metal contents. A mica layered with films of TiO₂ can be used herein.

[0051] The mica useful herein may comprise an additional surface treatment to reduce interactions with other materials, to modify the skin feel, and/or to improve dispersibility. For example, the surface may be hydrophobically modified to increase suspension in a hydrophobic phase or composition. The surface-treated interference pigments may have a coating comprising from about 0.1% to about 20%, alternatively from about 0.1% to about 10%, and alternatively from about 0.1% to about 3% of the total particulate weight. Nonlimiting examples of surface treatments useful herein include silicones, acrylate silicone copolymers, acrylate polymers, alkyl silane, isopropyl titanium tri-isostearate, sodium stearate, magnesium myristate, perfluoroalcohol phosphate, perfluoropolymethyl isopropyl ether, lecithin, carnauba wax, polyethylene, chitosan, plant lipid extracts and mixtures thereof, preferably, silicones, silanes, stearates and mixtures thereof.

Lauroyl Lysine

[0052] The composition can comprise lauroyl lysine at a level of from about 0.1% to about 10%, alternatively from about 0.3% to about 7%, alternatively from about 0.5% to about 5%.

[0053] The shape of the lauroyl lysine can be anything. In some examples, the shape of the lauroyl lysine can be spherical, flat polygonal (selected from flat pentagonal, flat hexagonal, and flat heptagonal, preferably flat hexagonal shape), or a combination thereof. Alternatively, the shape of the lauroyl lysine can be flat polygonal.

[0054] The lauroyl lysine can have a flat polygonal shape with a mean particle size of from about 10 microns to about 40 microns, alternatively from about 15 microns to about 35 microns, alternatively from about 20 microns to about 30 microns, in view of improved sensory feeling, as too small particles may not provide enough bouncy/soft feel and too big particles may provide scrubbing-like feeling. Commercial example of such lauroyl lysine in the flat polygonal shape having such particle size is, for example, Amihope LL available from Ajinomoto, having a flat hexagonal shape and having a mean particle size of 20-30 microns.

[0055] The lauroyl lysine can have a spherical shape with a mean particle size of from about 1 microns to about 30 microns, alternatively from about 2 microns to about 20 microns, alternatively from about 3 microns to 8 microns.

[0056] Lauroyl lysine, especially when it's in flat polygonal shape, may be dispersed in the composition as a layered structure comprising some or several flat polygonal crystals. It is believed that the lauroyl lysine provides improved moisturizing/cushioning feel, compared to starch solid powders, cellulose solid powders.

[0057] Also, compared to hydrophilic solid polymeric powders such as starch solid powders and cellulose solid powders, it is believed that the lauroyl lysine provides reduced pilling and/or clumping especially when the com-

position contains higher levels of hydrophilic thickeners. For such reduced pilling and/or clumping, it may be preferred that the composition is substantially free of such hydrophilic solid powders, i.e., contains 0.1% or less, alternatively 0.05% or less of such hydrophilic solid powders. Alternatively, the composition can be free of such hydrophilic solid powders, i.e., contains 0% of such hydrophilic solid powders.

[0058] It is also believed the lauroyl lysine may provide matte appearance and/or reduced visibility of skin pores.

Aqueous Carrier/Water

[0059] The composition can comprises an aqueous carrier. The aqueous carrier can be substantially water. Deionized water may be used. Water from natural sources including mineral cations can also be used, depending on the desired characteristic of the composition.

[0060] The composition can comprise water at a level of from about 55% to about 99%, alternatively from about 60% to about 98%, alternatively from about 60% to about 95%, alternatively from about 60% to about 90%, by weight of the composition.

[0061] In some instances, the compositions herein are in the form of an oil-in-water ("O/W") emulsion that provides a sensorial feel that is light and non-greasy. Suitable O/W emulsions herein may include a continuous aqueous phase of more than 50% by weight of the composition, and the remainder being the dispersed oil phase. The aqueous phase may include 1% to 99% water, based on the weight of the aqueous phase, along with any water soluble and/or water miscible ingredients. In these instances, the dispersed oil phase will typically be present at less than 30% by weight of composition (e.g., 1% to 20%, 2% to 15%, 3% to 12%, 4% to 10%, or even 5% to 8%) to help avoid some of the undesirable feel effects of oily compositions. The oil phase may include one or more volatile and/or non-volatile oils (e.g., botanical oils, silicone oils, and/or hydrocarbon oils).

pH of the Composition

[0062] Typically, cosmetic compositions are formulated to have a slightly acidic to neutral pH (i.e., 5.0-7.0), which is believed to improve the stability of certain ingredients in the composition (e.g., niacinamide, salicylates, and neutralized thickeners). However, formulating a skin care composition at low pH (e.g., 2.0-5.0) may also provide certain benefits such as improving the appearance of skin, bolstering the acid mantle of the skin, exfoliating the skin, improving skin texture, and/or providing flexibility in product formulation.

[0063] The composition may have a lower pH, i.e., pH of less than 5.0, alternatively from about 1.5 to about 4.9, alternatively from about 2.0 to about 4.5, alternatively from about 2.5 to about 4.0, or alternatively from about 3.5 to about 4.0.

Substantially Free of Microplastic Solid Particulates

[0064] The composition can contain lauroyl lysine can provide benefits from mica with reduced friction/reduced draggy feeling, without the use of microplastic solid particulates. Thus, the composition may also provide environmental benefit in view of the reduction of the use of microplastic solid particulates. Such microplastic solid particulates are, for example, nylon powder, polyurethane powder, polyethylene powder, silicone resin powder. The com-

position may be substantially free of such microplastic solid particulates. Alternatively, the composition may be free of such microplastic solid particulates, i.e., contains 0% of such microplastic solid particulates.

Skin Conditioning Agent

[0065] Optionally, the composition can further comprise a skin conditioning agent. These agents may be selected from humectants and emollients. The amount of skin-condition agent may range from about 1% to about 50%, alternatively from about 2% to about 40%, alternatively from about 5% to about 30%, by weight of the composition.

[0066] Humectants are polyhydric alcohols intended for moisturizing, reducing scaling and stimulating removal of built-up scale from the skin. Typical polyhydric alcohols include polyalkylene glycols and alternatively alkylene polyols and their derivatives. Illustrative are propylene glycol, dipropylene glycol, polypropylene glycol, polyethylene glycol, sorbitol, hydroxypropyl sorbitol, hexylene glycol, 1,3-butyleneglycol, 1,2,6-hexanetriol, ethoxylated glycerin, propoxylated glycerin and mixtures thereof. In some examples, the humectant can be glycerin.

[0067] When the conditioning agent is an emollient it may be selected from hydrocarbons, fatty acids, fatty alcohols and esters.

Fatty Alcohol

[0068] The compositions herein may include a fatty alcohol. Fatty alcohols refer to high-molecular-weight, straight-chain primary alcohols that have the general structure:



where $n=8$ to 32.

[0069] Fatty alcohols may be natural or synthetic, saturated or unsaturated, branched or straight-chain. Some non-limiting examples of fatty alcohols commonly used in cosmetic compositions include caprylic, capryl, lauryl, myristyl, cetyl, stearyl, and behenyl alcohols. The fatty alcohols herein may be referred to generically by the number of carbon atoms in the molecule. For example, a “C12 alcohol” refers to an alcohol that has 12 carbon atoms in its chain (i.e., dodecanol).

[0070] The fatty alcohol may be included in the compositions herein at 0.0001% to 15% (e.g., 0.0002% to 10%, 0.001% to 15%, 0.025% to 10%, 0.05% to 7%, 0.05% to 5%, or even 0.1% to 3%) by weight of the composition.

Whitening Agents

[0071] The present compositions may contain a whitening agent. The whitening agent useful herein refers to active ingredients that not only alter the appearance of the skin, but further improve hyperpigmentation as compared to pre-treatment. Useful whitening agents useful herein include ascorbic acid compounds, vitamin B 3 compounds, azelaic acid, butyl hydroxy anisole, gallic acid and its derivatives, hydroquinoline, kojic acid, arbutin, mulberry extract, tetrahydrocurcumin, and mixtures thereof. Use of combinations of whitening agents is also believed to be advantageous in that they may provide whitening benefit through different mechanisms.

[0072] When used, the compositions can contain from about 0.1% to about 10%, alternatively from about 0.2% to about 5%, by weight of the composition, of a whitening agent.

[0073] Ascorbic acid compounds are useful whitening agents and ascorbyl glucoside can be a preferred derivative of the ascorbic acid compounds.

Conditioning Agents

[0074] The compositions herein may include 0.1% to 50% by weight of a conditioning agent (e.g., 0.5% to 30%, 1% to 20%, or even 2% to 15%). Adding a conditioning agent can help provide the composition with desirable feel properties (e.g., a silky, lubricious feel upon application). Some non-limiting examples of conditioning agents include, hydrocarbon oils and waxes, silicones, fatty acid derivatives, cholesterol, cholesterol derivatives, diglycerides, triglycerides, vegetable oils, vegetable oil derivatives, acetoglyceride esters, alkyl esters, alkenyl esters, lanolin, wax esters, beeswax derivatives, sterols and phospholipids, salts, isomers and derivatives thereof, and combinations thereof.

[0075] Particularly suitable examples of conditioning agents include non-volatile silicone fluids, such as dimethicone copolyol, dimethylpolysiloxane, dimethylpolysiloxane, mixed C1-30 alkyl polysiloxanes, phenyl dimethicone, dimethiconol, dimethicone, dimethiconol, silicone crosspolymers, and combinations thereof. Dimethicone may be especially suitable, since some consumers associate the feel properties provided by certain dimethicone fluids with good moisturization.

Hydrophilic Thickener

[0076] The compositions, in some embodiments, may further include one or more hydrophilic thickeners. When present, the composition can include from about 0.01% to about 5%, alternatively from about 0.1% to about 4%, and alternatively from about 0.1% to about 3%, by weight of the composition of the hydrophilic thickeners.

[0077] The hydrophilic thickener useful herein is not particularly limited as long as it is one that is normally used in cosmetic products. Examples include natural or semi-synthetic water-soluble polymers, synthetic water-soluble polymers and inorganic water-soluble polymers.

[0078] As the natural or semi-synthetic water-soluble polymers, polysaccharides and derivatives thereof (including water-soluble alkyl-substituted polysaccharide derivatives) can be used. Specific examples can include plant-based polymers such as gum arabic, tragacanth gum, galactan, guar gum, carob gum, karaya gum, carrageenan, pectin, agar, quince seed (marmelo), algecolloid (phacophyceae extract), starch (rice, corn, potato, wheat) and glycyrrhizic acid; microbe-based polymers such as xanthan gum, dextran, succinoglycan and pullulan; starch-based polymers such as carboxymethyl starch and methylhydroxypropyl starch; cellulose-based polymers such as methyl cellulose, nitrocellulose, ethyl cellulose, methyl hydroxypropyl cellulose, hydroxyethyl cellulose, sodium cellulose sulfate, hydroxypropyl cellulose, sodium carboxymethyl cellulose (CMC), crystalline cellulose and cellulose powder; and alginic acid-based polymers such as sodium alginate and propylene glycol esters of alginic acid.

[0079] The synthetic water-soluble polymers include ionic or non-ionic water-soluble polymers, for example, vinyl-

based polymers such as polyvinyl alcohol, polyvinyl methyl ether, polyvinyl pyrrolidone and carboxyvinyl polymers (carbomers); acryl-based polymers such as sodium polyacrylate, poly ethyl acrylate, polyacrylamide compounds and acrylic acid/alkyl methacrylate copolymers (product name "pemulen TR-1").

[0080] The polyacrylamide compounds particularly include polyacrylamide compounds consisting of homopolymers, copolymers or crosspolymers containing one or more constituent units chosen from among 2-acrylamido-2-methylpropane sulfonic acid (hereinafter sometimes abbreviated to "AMPS"), acrylic acid and derivatives thereof.

[0081] Specific examples of such polyacrylamide compounds include vinylpyrrolidone/2-acrylamido-2-methylpropane sulfonic acid (salt) copolymers, dimethylacrylamide/2-acrylamido-2-methylpropane sulfonic acid (salt) copolymers, acrylamide/2-acrylamido-2-methylpropane sulfonic acid copolymers, dimethylacrylamide/2-acrylamido-2-methylpropane sulfonic acid crosspolymers cross-linked with methylenebisacrylamide, mixtures of polyacrylamide and sodium polyacrylate, sodium acrylate/2-acrylamido-2-methylpropane sulfonic acid copolymers, hydroxyethyl acrylate/2-acrylamido-2-methylpropane sulfonic acid (salt) copolymers, ammonium polyacrylate, polyacrylamide/ammonium acrylate copolymers, and acrylamide/sodium acrylate copolymers. However, the compounds are not limited to these examples.

[0082] Examples of salts in the previous paragraph can include alkali metal salts (such as calcium salts and magnesium salts), ammonium salts, organic amine salts (such as monoethanolamine salts, diethanolamine salts, and triethanolamine salts). One or more of these polyacrylamide compounds may be used.

[0083] These polyacrylamide compounds may be synthesized or obtained as commercial products. For example, the vinyl pyrrolidone/2-acrylamido-2-methylpropane sulfonic acid (salt) copolymer may be "Aristoflex AVC" (manufactured by Clariant), the sodium alkylate/2-acrylamido-2-methylpropane sulfonic acid (salt) copolymer may be "Simulgel EG" (manufactured by Seppic) or "Simulgel EPG" (manufactured by Seppic), the acrylamide/2-acrylamido-2-methylpropane sulfonic acid sodium salt copolymer may be "Simulgel 600" (manufactured by Seppic), the acrylamide/2-acrylamido-2-methylpropane sulfonic acid (salt) may be "Sepigel 305" (manufactured by Seppic) or "Sepigel 501" (manufactured by Seppic), the homopolymer of a 2-acrylamido-2-methylpropane sulfonic acid sodium salt may be "1-lostacerin AMPS" (manufactured by Clariant) or "Simulgel 800" (manufactured by Seppic), and the dimethylacrylamide/2-acrylamido-2-methylpropane sulfonic acid may be "SU-Polymer 0-1" (manufactured by Toho Chemical Industry).

[0084] It may be preferred to use: carboxyvinyl polymers (carbomers), for example, those having tradenames of Carbopol 940 and Carbopol 980; and acrylic acid/alkyl (meth) acrylate copolymers, for example that having an INCI name of Acrylates/C10-30 Alkyl Acrylate Crosspolymer, for example, those having tradenames of Carbopol 1342 Polymer, Carbopol Ultrez 20, Carbopol Ultrez 21, Pemulen TR-1, and Pemulen TR-2. By the use of lauroyl lysine together with these thickeners, the composition provides reduced stickiness and/or draggy feel coming from these thickeners especially when using such polymers at a higher

level such as 0.2% or more, while having a sufficient thickening efficacy from these thickeners.

[0085] The hydrophilic thickener in the water-based cosmetic may be a combination of one or more types.

Emulsifiers

[0086] When the composition is in the form of an emulsion, it may contain an emulsifier. Emulsifiers may be nonionic, anionic, cationic, or zwitterionic. Some non-limiting examples of emulsifiers are disclosed in U.S. Pat. Nos. 3,755,560, 4,421,769, U.S. Publication No. 2006/0275237 and McCutcheon's Detergents and Emulsifiers, North American Edition, pages 317-324 (1986). Suitable examples of emulsifiers include non-ionic stearic acid derivative emulsifiers such as Polysorbate-20, glyceryl stearate, sorbitan stearate, PEG-100 stearate, Steareth-2, Steareth-21, glycereth-25 pyrrolidonecarboxylic acid isostearate and combinations of these.

Other Optional Ingredients.

[0087] Compositions herein may include one or more optional ingredients known for use in topical cosmetic compositions, provided that the optional components do not unacceptably alter the desired benefits of the composition. In some instances, it may be desirable to select cosmetic actives that function via different biological pathways so that the actives do not interfere with one another. When the composition is in the form of an emulsion, the additional ingredients should not introduce instability into the emulsion (e.g., syneresis). For example, it may be desirable to select optional ingredients that do not form complexes with other ingredients in the composition, especially pH sensitive ingredients like vitamin B3 compounds, salicylates and peptides.

[0088] The additional ingredients should be suitable for use in contact with human skin tissue without undue toxicity, incompatibility, instability, allergic response, and the like. The optional components, when present, may be included at an amount of about 0.001% to 50% (e.g., 0.01% to 40%, 0.1% to 30%, 0.5% to 20%, or 1% to 10%), by weight of the composition. Some nonlimiting examples of additional ingredients include vitamins, minerals, peptides and peptide derivatives, sugar amines, sunscreens, oil control agents, particulates, flavonoid compounds, hair growth regulators, anti-oxidants and/or anti-oxidant precursors, preservatives, protease inhibitors, tyrosinase inhibitors, anti-inflammatory agents, moisturizing agents, exfoliating agents, skin lightening agents, sunscreen agents, sunless tanning agents, lubricants, anti-acne agents, anti-cellulite agents, chelating agents, anti-wrinkle actives, anti-atrophy actives, phytosterols and/or plant hormones, N-acyl amino acid compounds, antimicrobials, and antifungals. Some particularly suitable examples of additional ingredient include one or more skin care actives selected from the group consisting of vitamin B3 compounds (e.g., niacinamide), vitamin B5 compounds (e.g., d-panthenol), n-acyl amino acids (e.g., undecylenoyl phenylalanine), vitamin E compounds (e.g., tocopheryl acetate), palmitoylated dipeptides (e.g., palmitoyl-lysine-threonine), palmitoylated pentapeptides (e.g., palmitoyl-lysine-threonine-threonine-lysine-serine), vitamin A compounds (e.g., retinol and retinyl propionate), and combinations thereof. Other non-limiting examples of optional ingredients and/or skin care actives that may be

suitable for use herein are described in U.S. Publication Nos. 2002/0022040; 2003/0049212; 2004/0175347; 2006/0275237; 2007/0196344; 2008/0181956; 2008/0206373; 2010/0092408; 2008/0206373; 2010/0239510; 2010/0189669; 2010/0272667; 2011/0262025; 2011/0097286; US2012/0197016; 2012/0128683; 2012/0148515; 2012/0156146; and 2013/0022557; and U.S. Pat. Nos. 5,939,082; 5,872,112; 6,492,326; 6,696,049; 6,524,598; 5,972,359; and 6,174,533.

[0089] Sucrose esters may be used herein. Such sucrose ester can be a blend of two or more sucrose esters, wherein the two or more sucrose esters are present at a ratio of any one sucrose ester to another of 1:10 to 1:1 (e.g., 1:7, 1:5, 1:3, or 1:2). In some instances, the sucrose ester may be a blend of sucrose laurate and sucrose dilaurate, wherein sucrose laurate is present at 50% to 80%, by weight of the sucrose ester, and the sucrose dilaurate is present at 20% to 45%, by weight of the sucrose ester. Alternatively, the sucrose ester may be a blend of sucrose laurate, sucrose dilaurate and sucrose trilaurate, wherein the sucrose dilaurate is present at 35% or more, by weight of the sucrose ester. A particularly suitable example of a sucrose ester for use herein is Sucrose Dilaurate BC10034 available from BASF.

Method of Use of the Cosmetic Composition

[0090] The method of use herein includes identifying a target portion of skin on a person in need of treatment and applying the composition to the target portion of skin over the course of a treatment period. The target portion of skin may be on a facial skin surface such as the forehead, perioral, chin, periorbital, nose, and/or cheek) or another part of the body (e.g., hands, arms, legs, back, chest). The person in need of treatment is one whose skin exhibits signs of oxidative stress, such as fine lines, wrinkles, hyperpigmentation, uneven skin tone, and/or other visible skin features typically associated with aging. In some instances, the target portion of skin may not exhibit a visible sign of skin aging, but a user (e.g., a relatively young user) may still wish to target such an area of skin, if it is one that typically develops such issues as a person ages. In this way, the present method may be used as a preventative measure to delay the onset of visible signs of skin aging.

[0091] The composition may be applied to a target portion of skin and, if desired, to the surrounding skin at least once a day, twice a day, or on a more frequent daily basis, during a treatment period. When applied twice daily, the first and second applications are separated by at least 1 to 12 hours. Typically, the composition is applied in the morning and/or in the evening before bed. The treatment period may last for at least 1 week (e.g., about 2 weeks, 4 weeks, 8 weeks, or even 12 weeks). In some instances, the treatment period will extend over multiple months (i.e., 3-12 months). In some instances, the composition may be applied most days of the week (e.g., at least 4, 5 or 6 days a week), at least once a day or even twice a day during a treatment period of at least 2 weeks, 4 weeks, 8 weeks, or 12 weeks.

[0092] The step of applying the composition may be accomplished by localized application. In reference to application of the composition, the terms "localized", "local", or "locally" mean that the composition is delivered to the targeted area (e.g., a wrinkle or line) while minimizing delivery to skin surfaces where treatment is not desired. The composition may be applied and lightly massaged into an area of skin. The form of the composition or the dermatology

logically acceptable carrier should be selected to facilitate localized application. While certain embodiments herein contemplate applying a composition locally to an area, it will be appreciated that compositions herein can be applied broadly to one or more skin surfaces. In certain embodiments, the compositions herein may be used as part of a multi-step beauty regimen, wherein the present composition may be applied before and/or after one or more other compositions.

Combinations

- [0093]** A cosmetic composition comprising by weight:
- [0094]** from about 0.01% to about 5% of mica;
- [0095]** from about 0.1% to about 10% of lauroyl lysine; and
- [0096]** from about 55% to about 99% of water;
- [0097]** wherein the composition meets at least one of the following conditions (i)-(iii):
- (i) wherein the weight ratio between the mica and lauroyl lysine is from about 2:1 to about 1:10;
- (ii) wherein the mica has a mean particle size of from about 5 microns to about 20 microns, and the lauroyl lysine has a mean particle size of from about 10 microns to about 40 microns, and the mean particle size ratio between the mica and lauroyl lysine is from about 1:1 to about 1:5; and
- (iii) wherein the lauroyl lysine has a flat polygonal shape.
- [0098]** The composition of the preceding feature, wherein the weight ratio between the mica and lauroyl lysine is from about 1:1 to about 1:7, preferably from about 1:1.2 to about 1:7, ore preferably from about 1:1.5 to about 1:5.
- [0099]** The composition of any of the preceding features, wherein the mean particle size ratio between the mica and lauroyl lysine is from about 1:1.2 to about 1:4, preferably from about 1:1.4 to about 1:4.
- [0100]** The composition of any of the preceding features, wherein the composition comprises from about 0.05% to about 3% of mica, preferably from about 0.1% to about 2% of mica.
- [0101]** The composition of any of the preceding features, wherein the composition comprises from about 0.3% to about 7% of lauroyl lysine, preferably from about 0.5% to about 5% of lauroyl lysine.
- [0102]** The composition of any of the preceding features, further comprising up to about 0.5%, preferably up to about 0.3%, alternatively up to about 0.1% of a total level of cetyl, stearyl, and behenyl alcohols.

[0103] The composition of any of the preceding features, wherein the mica is layered with films of TiO₂.

[0104] Use of the composition of any of the preceding features, for reducing friction or draggy feeling.

[0105] Use of the composition of any of the preceding features, for providing the reduction of the use of micro-plastic solid particulates, as an environmental benefit.

EXAMPLES

Cosmetic Compositions (wt %)

	CEx. i	Ex. 1	Ex. 2
Water Purified	q.s. to 100%	q.s. to 100%	q.s. to 100%
Glycerin	4.0	4.0	4.0
Xylitol	0.5	0.5	0.5
Trehalose			
Aminomethyl propanol	1.3	1.3	1.3
Sodium Benzoate	0.05	0.05	0.05
Disodium EDTA	0.1	0.1	0.1
Pentylene Glycol	2.0	2.0	2.0
1,3-Butylene Glycol	5.0	5.0	5.0
Benzyl Alcohol	0.2	0.2	0.2
Triethylhexanoin	3.5	3.5	3.5
Caprylic Capric Triglyceride	0.5	0.5	0.5
Eldew PS-203 *1			
PMX-200 Silicone Fluid 5cSt *2			
DC1503 *3			
Boron Nitride	0.8	0.8	0.8
Synthetic Mica and TiO ₂ *4	1.0	1.0	1.0
Natural Mica and TiO ₂ *5			
Lauroyl Lysine *6		2.0	4.0
Polysorbate 20	0.25	0.25	0.25
PEG-32*7	1.0	1.0	1.0
Sepigel 305 *8	0.5	0.5	0.5
Ultrez 20 *9	0.26	0.26	0.26
Pemulen TR-2 *10			
Xanthan gum	0.05	0.05	0.05
Niacinamide	5.0	5.0	5.0
Lactic Acid			
Sodium Lactate			
Phenoxyethanol	0.25	0.25	0.25
Perfume	0.03	0.03	0.03

Details of the Ingredients Used in the Above Table

*1	Eldew PS-203	INCI: Phytosteryl/Octyldodecyl Lauroyl Glutamate
*2	PMX-200 Silicone Fluid 5 cSt	Dimethicone 5 cSt
*3	DC1503	Dimethicone (5 cSt, Solvent) and Dimethiconol
*4	Synthetic Mica and TiO ₂	Synthetic Mica layered with films of TiO ₂ , wherein the Synthetic Mica has a mean particle size of 10-20 microns
*5	Natural Mica and TiO ₂	Natural Mica layered with films of TiO ₂ , wherein the Natural Mica has a mean particle size of 10-20 microns
6	Lauroyl Lysine	having a mean particle size of 20-30 microns and having a flat hexagonal shape.
*7	PEG-32	having MW of 1540
*8	Sepigel 305	Containing 40% of Polyacrylamide as active. INCI: Polyacrylamide & Water & C13-14 Isoparaffin & Laureth-7
*9	Ultrez 20	INCI: Acrylates/C10-30 Alkyl Acrylate Crosspolymer
*10	Pemulen TR-2	INCI: Acrylates/C10-30 Alkyl Acrylate Crosspolymer

Heavy Coated Feel and Higher Friction/Draggy Feel

[0106] Heavy coated feel and Higher friction/Draggy feel are evaluated by the following method: Sensory Expert Grading Method (DAP Sensory Measurement)

[0107] Blinded, monadic, single-use evaluations are conducted under controlled environmental conditions (20-25° C./45-50% RH), by 12 trained expert panels using standardized procedure. Evaluations are done on one side of cheeks. 0.1 ml of a sample is applied to a washed cheek. Using the pads of the index and middle finger of the dominant hand, the Sample is applied with 15 circular rotations about 2 inches in diameter) in the center of the cheek. Tactile feel of the sample is evaluated using the absolute standard score from 0 to 8. Tactilecoate (perception of the amount of residue on skin) and Swipedrag (perception of resistance to movement) of a sample are evaluated after 10 min and 15 min of sample application, respectively. Higher number in Tactilecoate means the sample has more residual feel on skin, which is associated with heavy coated feel. Higher number in Swipedrag means the sample is more resistant to spread over the skin, which is associated with higher friction/draggy feel.

[0108] Each panelist records its data on a hard-copy pre-coded ballot. Each test site is washed and equilibrated between test samples.

[0109] The results of some of the above compositions are shown below in Table.

		Result		
		CEx. i	Ex. 1	Ex. 2
After 10 min	Heavy coated feel (tactilecoate)	3.7	3.5	3.2
	higher friction/Draggy feel (swipedrag)	4	3.7	3.4
After 15 min	Heavy coated feel (tactilecoate)	3.2	3.1	2.8
	higher friction/Draggy feel (swipedrag)	3.8	3.4	3.2

[0110] CEx. i is a comparative examples that do not contain lauroyl lysine. Ex. 1 and Ex. 2 are inventive examples, which are suitably used as an oil-in-water emulsion cosmetic composition, in a form of essence, lotion, serum and/or gel cream. Ex. 1-2 provide reduced heavy coated feel, and reduced friction/reduced draggy feel, as compared to the comparative example.

[0111] The dimensions and values disclosed herein are not to be understood as being strictly limited to the exact numerical values recited. Instead, unless otherwise specified, each such dimension is intended to mean both the recited value and a functionally equivalent range surrounding that value. For example, a dimension disclosed as “40 mm” is intended to mean “about 40 mm”.

[0112] Every document cited herein, including any cross referenced or related patent or application and any patent application or patent to which this application claims priority or benefit thereof, is hereby incorporated herein by reference in its entirety unless expressly excluded or otherwise limited. The citation of any document is not an admission that it is prior art with respect to any invention disclosed or claimed herein or that it alone, or in any combination with any other reference or references, teaches, suggests or discloses any such invention. Further, to the extent that any

meaning or definition of a term in this document conflicts with any meaning or definition of the same term in a document incorporated by reference, the meaning or definition assigned to that term in this document shall govern.

[0113] While particular embodiments of the present invention have been illustrated and described, it would be obvious to those skilled in the art that various other changes and modifications can be made without departing from the spirit and scope of the invention. It is therefore intended to cover in the appended claims all such changes and modifications that are within the scope of this invention.

What is claimed is:

1. A cosmetic composition comprising by weight:

from about 0.01% to about 5% of mica;

from about 0.1% to about 10% of lauroyl lysine; and

from about 55% to about 99% of water;

wherein the composition meets at least one of the following conditions (i)-(iii):

(i) wherein the weight ratio between the mica and lauroyl lysine is from about 2:1 to about 1:10;

(ii) wherein the mica has a mean particle size of from about 5 microns to about 20 microns, and the lauroyl lysine has a mean particle size of from about 10 microns to about 40 microns, and the mean particle size ratio between the mica and lauroyl lysine is from about 1:1 to about 1:5; and

(iii) wherein the lauroyl lysine has a flat polygonal shape.

2. The composition of claim 1, wherein the weight ratio between the mica and lauroyl lysine is from about 1:1 to about 1:7.

3. The composition of claim 2, wherein the weight ratio between the mica and lauroyl lysine is from about 1:1.2 to about 1:7.

4. The composition of claim 3, wherein the weight ratio between the mica and lauroyl lysine is from about 1:1.5 to about 1:5.

5. The composition of claim 1, wherein the mean particle size ratio between the mica and lauroyl lysine is from about 1:1.2 to about 1:4.

6. The composition of claim 5, wherein the mean particle size ratio between the mica and lauroyl lysine is from about 1:1.4 to about 1:4.

7. The composition of claim 1, wherein the composition comprises from about 0.05% to about 3% of mica.

8. The composition of claim 7, wherein the composition comprises from about 0.1% to about 2% of mica.

9. The composition of claim 1, wherein the composition comprises from about 0.3% to about 7% of lauroyl lysine.

10. The composition of claim 9, wherein the composition comprises from about 0.5% to about 5% of lauroyl lysine.

11. The composition of claim 1, further comprising up to about 0.5% of a total level of cetyl, stearyl, and behenyl alcohols.

12. The composition of claim 11, further comprising up to about 0.3% of a total level of cetyl, stearyl, and behenyl alcohols.

13. The composition of claim 12, further comprising up to about 0.1% of a total level of cetyl, stearyl, and behenyl alcohols.

14. The composition of claim 1, wherein the mica is layered with films of TiO₂.

* * * * *